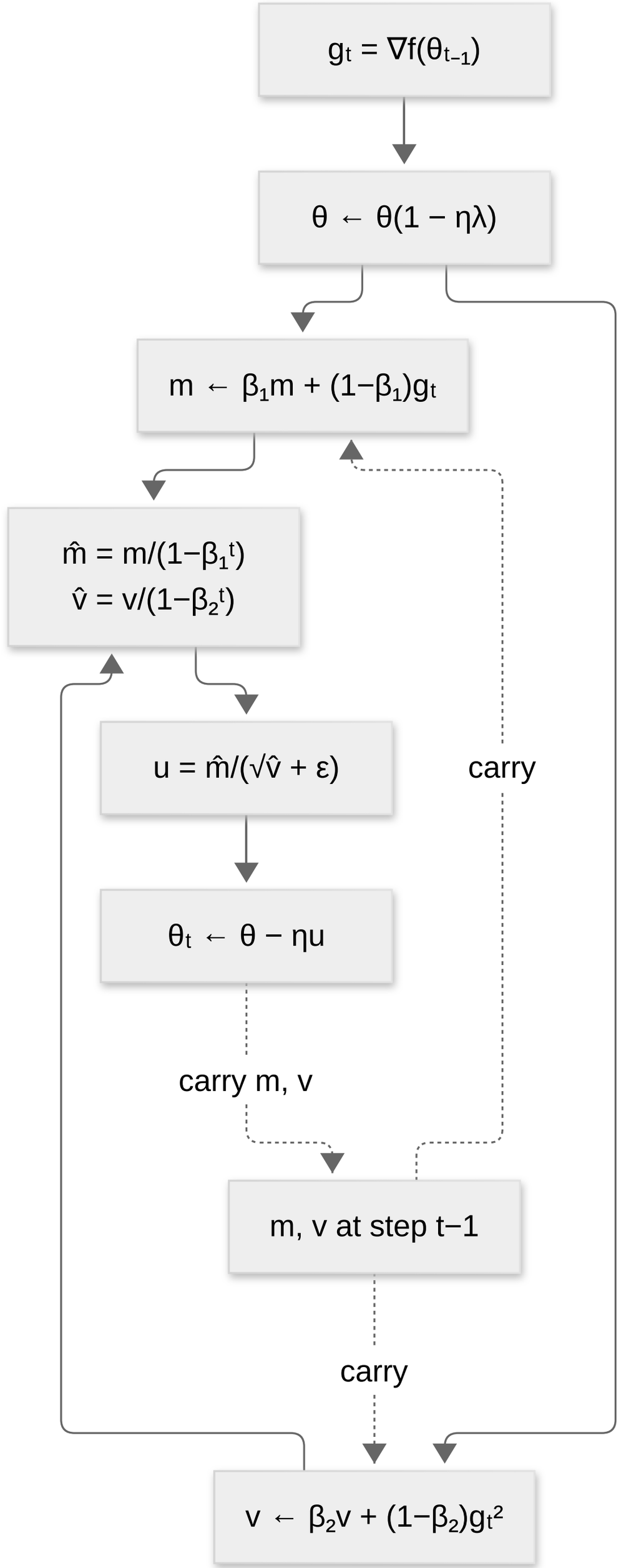

$g_t = \nabla f(\theta_{t-1})$
$\theta \leftarrow \theta(1 - \eta\lambda)$
$m \leftarrow \beta_1 m + (1-\beta_1) g_t$
$\hat{m} = m/(1-\beta_1^t)$
$\hat{v} = v/(1-\beta_2^t)$
$u = \hat{m}/(\sqrt{\hat{v}} + \varepsilon)$
carry
$\theta_t \leftarrow \theta - \eta u$
carry m, v
m, v at step t−1
carry
$v \leftarrow \beta_2 v + (1-\beta_2) g_t^2$